



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 31.07.2026

In addition to Part I (General Handout for all courses appended to the time table) this portion further gives specific details regarding the course.

Course No. : MF F316
Course Title : Machining and Machine Tools
Instructor-in-Charge : Prof. Tribeni Roy
Lab Instructor : Ms. Astha Pujara

1. Course Description:

Significance of metal machining and machine tools, Geometry of cutting tools, Mechanics of machining processes and analysis, Role of cutting fluids and properties, Abrasive machining processes and analysis, Economics of machining, Regulation of speed and feed rates in machine tools, Design of machine tool structures, Guideways, Spindles, Design of cutting tools for various machine tools, Jigs and Fixtures

2. Scope and Objective of the Course:

This course is designed to enrich theoretical, analytical as well as practical knowledge about metal cutting techniques and various machine tools used in manufacturing. The course includes understanding of important factors associated with machining such as what is the mechanics of cutting, why cutting fluid is necessary in metal removal, and how abrasive machining processes are different from tool-based cutting processes. The course also includes design aspect of cutting tools, machine parts (structure, spindles, jigs-fixtures) would be explored to comprehend the knowledge of machine tools.

3. Text Books:

i) **A. B. Chattopadhyay**, Machining and machine tools, 2nd edition, Wiley Publishers

4. Reference books

- i) **A. Ghosh and A. K. Mallik**, Manufacturing Science, East-West Press Private Limited
- ii) **N K Mehta**, Machine tool design and NC control, third edition, Tata McGraw-hill
- iii) **N K Mehta**, Metal Cutting and Design of Cutting Tools, Jigs & Fixtures, Tata McGraw-hill.
- iv) **G. K. Lal**, "Introduction to machining Science" Third edition, New Age International Publishers.
- v) **S. Kalpakjian and S.R. Schmid**, "Manufacturing Processes for Engineering Materials", Pearson Publications, Fifth Edition
- vi) **P. N. Rao**, "Manufacturing technology Volume II Metal Cutting and Machine Tools" McGraw Hill Education (India) Private Limited





5. Course Plan:

Module No.	Lecture session	References	Outcome(s)
1	L1 – L15: Introduction to machining, understanding tool geometry, chip formation & heat generation during cutting	TB1, RB1	Understanding the significance of machining, basic machining processes and machine tools. Analyse the chip formation process during machining and relevant heat generation in the process
2	L16 – L20: Tool life and machinability	TB1, RB1	To understand the tool wear and effects of different parameters on tool life. To compute the various forces and power consumption during turning and boring operations. Also, to compute the thrust and torque during a drilling/milling process.
3	L21 – 23: Machine tools, abrasive machining processes and analysis.	TB1, RB1	To demonstrate basic elements, components, power transmission, actuation systems and guide ways of machine tools. To realize forces and energy consumption during grinding operations. To understand the selection of grinding wheels.
4	L24 – L30: Economics of machining and regulation of speed and feed rates in machine tools	RB1, RB4	Economics of machining in terms of machining cost. Optimization of cutting speed for min. cost, max. production rate and maximum profit. Designing the speed gear box for any machine tool, Decide the range and number of speeds. Appropriate gear layout and optimum ray diagram.
5	L31 – L35: Design of machine tool structures, Guideways, Spindles, Design of cutting tools for various machine tools, Jigs and Fixtures	TB1, RB3	Determining the characteristics of machine tool structures using concepts of solid mechanics. Determination of overall dimension of cutting tools and their mounting features. Also, placing, positioning, and orienting the workpiece using jig and fixtures.





6. Lab Practical:

Various experimentations as part of this course will be conducted via demonstration/hands-on mode.

7. Evaluation Scheme:

Components	Duration (minutes)	Weightage (%)	Date	Remarks
Mid-semester test	90	30	07/10 AN1	Open Book
Assignments/Projects/Seminars/Quiz	--	20	To be announced	CB/OB
Lab quiz/viva/assignment	--	15	To be announced	CB/OB
Comprehensive Examination	180	35	10/12 FN	Closed Book

8. Chamber Consultation Hours:

To be announced in the first class.

9. Attendance Policy:

Each student is expected to attend all classes and participate in all evaluation components.

10. Notices:

All notices related to this course will be communicated via email.

11. Make-up Policy:

Make-up will be granted ONLY in genuine cases with prior permission.

12. Minimum Passing Criteria: The student who won't attend mid-semester /comprehensive examination will be awarded NC. Further, the passing marks for this course will be determined based on one of the two criteria: *30% of the median of the post-comprehensive examination score or 20% of the average of the first three top scorers. The lower value between these two criteria will be considered as a passing mark in this course.*

Instructor- in-Charge
MF F316

